Johannes Hosle

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EMPLOYMENT

Carnegie Mellon University in Qatar 2026–present
Postdoctoral Fellow in Mathematics
Mentors: Layan El Hajj and Henrik Shahgholian

EDUCATION

Massachusetts Institute of Technology 2020–2026
Ph.D. in Mathematics
Advisor: David Jerison

University of California, Los Angeles 2016–2020
B.Sc. in Mathematics with Departmental Honors, *summa cum laude*
M.A. in Mathematics

PUBLICATIONS AND PREPRINTS

J. Hosle, Monotonicity formulas in positively curved settings with applications to two-phase free boundary problems, submitted.

J. Hosle, The isomorphic section-projection problem for convex bodies, submitted.

J. Hosle and P. Ivanisvili, Geometric Block Exponents and a Uniform Mixed-Alphabet Sumset Inequality, submitted.

J. Hosle, Stolarsky-Type Inequalities in a Max-Convolution Problem, submitted.

J. Hosle, A. V. Kolesnikov, and G. V. Livshyts, On the L_p-Brunn–Minkowski and dimensional Brunn–Minkowski conjectures for log-concave measures, *J. Geom. Anal.* (2021).

J. Hosle, On extensions of the Loomis–Whitney inequality and Ball's inequality for concave, homogeneous measures, *Adv. Appl. Math.* (2020).

J. Hosle, On the Comparison of Measures of Convex Bodies via Projections and Sections, *Int. Math. Res. Not. IMRN* (2021).

HONORS AND AWARDS

Sigma Xi 2026
Dean of Science Fellowship, MIT 2020–2023
Phi Beta Kappa 2020
Sherwood Prize, UCLA 2020
Dean's Honors List 2016–2020
Putnam Competition, Top 250 2016
National Merit Scholarship Semifinalist 2015

TEACHING AND GRADING

18.095 Mathematics Lecture Series, MIT Jan. 2026
Teaching Assistant (J. Dunkel)

18.065 Matrix Methods in Data Analysis, Signal Processing, and Machine Learning, MIT Spring 2025
Teaching Assistant (Z. Chen)

18.100B Real Analysis, MIT Fall 2024
Course Assistant (L. Guth)

18.100P Real Analysis, MIT Spring 2024
Recitation Leader (K. Naff)

18.01 Calculus, MIT Fall 2023
Recitation Leader (L. Guth)

MENTORSHIP EXPERIENCE

Academic Mentor, Mathroots @ MIT July 2026
Two-week mathematical talent accelerator for high school students hosted by MIT PRIMES.

MIT Directed Reading Program Jan. 2025
Mentored undergraduates Ryan Dong and Caleb Pascale in convex geometry, including the Busemann–Petty problem and Bourgain's bound for the isotropic constant.

PRESENTATIONS

Analysis Seminar, Columbia University Sept. 2026
Upcoming

Analysis Seminar, Brown University Sept. 2026
Upcoming

Synergies between Geometry, Probability, and Computation in High Dimensions Aug. 2026
ICERM, Brown University, Providence, Rhode Island

Workshop in PDE and Applied Math in the Northeast Region Mar. 2026
University of Connecticut, Storrs, Connecticut

Pure Math Graduate Student Seminar, MIT Feb. 2025

BIRS Workshop: Geometric Tomography Feb. 2020
BIRS, Banff, Canada

Workshop in Convexity and Geometric Aspects of Harmonic Analysis Dec. 2019
Georgia Institute of Technology, Atlanta, Georgia

RESEARCH PROGRAMS AND SUMMER SCHOOLS

Interactions between Convex Geometry and Spectral Analysis July 2025
Université de Montréal, Montréal, Canada

Particle Interactive Systems: Analysis and Computational Methods June 2024
SLMath, Berkeley, California

Georgia Tech REU in Mathematics Summer 2018
Supervisors: Michael Lacey and Galyna Livshyts

PROGRAMMING

Python: Proficient; background in statistical learning theory; experience with NumPy and scikit-learn.

Julia: Proficient; used for gradient flows in free boundary problems, supersolution construction, and numerical estimates for weighted Wirtinger inequalities.

Wolfram Language: Intermediate; familiar with machine learning methods.

LaTeX: Proficient.

LANGUAGES AND CITIZENSHIP

Languages: English (native), Russian (elementary).

Citizenships: United States and Germany.